Battery recycling equipment at 1–5 t/h: a buyer's guide for CAPEX-first decisions

BHS-Sonthofen remains the safest, most proven choice for mechanical black mass production — but it is no longer the only strong option, and for a CAPEX-first buyer, it may not be the optimal one. **The ANDRITZ/Duesenfeld partnership now offers a credible alternative that could cut plant costs by roughly 50%** through its low-temperature vacuum process, which eliminates gas scrubbing and stainless steel equipment entirely. Duesenfeld Meanwhile, Chinese manufacturers like Genox and Xingmao have demonstrated working international installations at the 1–2 t/h scale for **$500K–$1.5M** — a fraction of the €8–20M a BHS line commands. The real question is not which equipment is "best" in the abstract but which trade-offs align with the buyer's regulatory environment, risk tolerance, and CAPEX ceiling. This report provides a decision framework built on current evidence.

BHS-Sonthofen is proven but no longer the only credible option

BHS-Sonthofen's position as the leading Western supplier for battery mechanical recycling is well-earned. Its VR-series rotary shear and NGU-series universal shredder, combined with the proprietary HTC vacuum dryer (units delivered at **20,000-liter capacity**), Recovery-worldwide form a gas-tight process chain from shredding through to dried black mass. The company has now delivered **three large-scale commercial plants**, including the BASF Schwarzheide facility in Germany (15,000 t/year, operational since 2024) and two other undisclosed European installations. Recovery rates exceed **97%** for cathode active materials. Bhs-sonthofen BHS also opened a Charlotte, NC office in July 2024, signaling North American expansion.

However, three developments since 2022 have eroded BHS's monopoly position at the 1–5 t/h scale. First, **Erdwich Zerkleinerungs-Systeme** (Germany) now explicitly targets the **50 kg/h to 5 t/h range** with both dry (N_2-inerted) and wet shredding options — a rare combination. Erdwich Erdwich Their twin-shaft shredders handle full EV packs up to 900 kg without prior disassembly. Erdwich Second, **ELDAN Recycling** (Denmark) launched the ELDAN TOWER, an integrated shredding system combining their Super Chopper and Multi Purpose Rasper in a single nitrogen-blanketed tower rated at **4 t/h input capacity**. Eldan Recycling Third, MTB Recycling (France) has been operating its ZWB pilot line at **1 t/h since September 2023**, with industrial-scale configurations available to 4 t/h. Mtb-recycling MTB's containerized Drycell Box system also offers rapid, modular deployment for production scrap processing. MTB Recycling

For a CAPEX-first buyer, BHS's premium German engineering and multi-year operational pedigree may be worth the investment if the project requires the lowest risk profile and maximum regulatory credibility. But the competitive field has meaningfully expanded.

The Duesenfeld process changes the CAPEX equation

The most consequential development for cost-conscious buyers is the formalized ANDRITZ/Duesenfeld partnership. Duesenfeld's patented process operates entirely below **80°C under vacuum** ANDRITZ LinkedIn

(≤300 hPa), FPO which creates three cascading cost advantages that no other Western process currently matches.

First, at temperatures below 80°C, **no hydrogen fluoride (HF) gas forms from PVDF binder decomposition**. This eliminates the need for expensive HF-rated gas scrubbing systems — a major line item in conventional plants. Second, without corrosive HF in the process, **standard carbon steel equipment replaces stainless steel** throughout the line, reducing material costs significantly. ANDRITZ Third, the vacuum process **recovers electrolyte at >99.5% purity** for resale — creating a revenue stream that conventional drying processes destroy by thermally decomposing the electrolyte. SlideShare Duesenfeld claims these factors "halve" plant and operating costs versus conventional approaches. Duesenfeld

ANDRITZ brings its global EPC capability ANDRITZ (29,000+ employees, €7B+ revenue) to deliver turnkey installations, while Duesenfeld provides the process license and IP. Their reference plant design operates at **2 t/h (≈13,750 t/year)** ANDRITZ — squarely in the target range. The process achieves **91% recycling rate at cell level**, Energybusinessreview exceeding EU Battery Regulation minimums.

The main trade-off: fewer large-scale commercial references than BHS. Duesenfeld has operated its Wendeburg pilot plant for years, but the number of fully commissioned, multi-year commercial installations remains smaller. A CAPEX-first buyer willing to accept marginally higher technology risk should seriously evaluate this pathway.

Chinese manufacturers offer 50–70% CAPEX savings with real trade-offs

The Chinese battery recycling equipment market has matured considerably since 2022, and dismissing it entirely would be a mistake for budget-constrained buyers. However, the landscape is uneven — a handful of credible suppliers exist alongside many generic machinery companies rebranding for the sector.

Genox Recycling Technology (Guangzhou) stands out as the most sophisticated Chinese option. Genox maintains partnerships with Sicon GmbH (Germany), GENOX UK, and GENOX USA, suggesting quality standards above typical Chinese competitors. Genoxtech Genoxtech Their lithium battery recycling line features **oxygen-controlled charged crushing** — meaning batteries can be processed without pre-discharge, a significant operational simplification. Genoxtech Recovery rates of approximately 95% are claimed. Genoxtech

Gongyi Xingmao Machinery provides the strongest verification of international deployment. Their documented installations include a **1.5 t/h line in the USA**, a **1 t/h line in Singapore**, and a **2 t/h plant in South Korea** — all operational. Lithiumbatterybroken These reference installations at the target scale are rare even among Western suppliers.

Henan Doing Environmental Protection Technology offers the most transparent pricing: **$70,000 to $350,000** for lines processing 500–3,000 kg/h, with a modular design approach. Copperwirerecyclingmachinery At the upper end, this represents roughly one-tenth the cost of a comparable BHS installation.

The aggregate CAPEX picture for a complete Chinese-sourced mechanical black mass line at 1–5 t/h is approximately **$500K–$1.5M**, versus **€8–20M ($9–22M) from Western suppliers**. The savings are dramatic but carry real risks:

- **No ATEX certification was found for any Chinese battery recycling line.** Black mass dust creates ATEX Zone 20/21/22 environments. (Palamatic Process) Operating uncertified equipment in the EU is a regulatory and insurance liability. In less-regulated jurisdictions, this constraint relaxes.
- **CE marking claims vary in rigor.** Chinese manufacturers commonly self-declare CE compliance, but third-party conformity assessment is rare. The quality of compliance documentation varies enormously.
- **After-sales support degrades with distance.** Spare parts lead times of 4–8 weeks from China, timezone barriers, and language challenges create real operational risk. Only Genox (with European/US offices) (Genoxtech) and Xingmao (with verified international installations) have demonstrated meaningful after-sales presence outside China.
- **Component longevity is the most frequently cited concern.** Shredder blade life, bearing quality, and control system reliability (Chinese PLCs vs. Siemens/ABB) affect total cost of ownership over a 10–15 year plant life.

A pragmatic hybrid approach merits consideration: **Western-grade equipment for safety-critical components** (shredders, N_2 systems, gas management, ATEX-rated electricals) paired with **Chinese equipment for separation stages** (screening, magnetic/eddy current separation, air classification) where safety stakes are lower.

Supplier	Origin	Scale	CAPEX estimate	Key strength	Key risk
BHS-Sonthofen	Germany	1–15 kt/yr	€8–20M	3+ proven plants, >97% recovery	Premium cost
ANDRITZ/Duesenfeld	Austria/Germany	2 t/h reference	Est. 40–50% less than BHS	No HF, no stainless, electrolyte recovery	Fewer commercial references
Erdwich	Germany	50 kg–5 t/h	Not published	Wet+dry options, exact scale match	No published battery references
MTB Recycling	France	1–4 t/h	Est. lower than BHS	Containerized Drycell Box, 1 t/h pilot running	Newer to batteries
ELDAN Recycling	Denmark	4 t/h	Not published	Integrated TOWER shredding system	Recently launched, unproven
Primobius/SMS	Germany	2,500 tpa	€18.8M (incl. hydromet)	Mercedes reference, wet shredding	Lower throughput (0.3 t/h)
Genox	China	Flexible	$500K–$1.2M	European partnerships, no pre-discharge needed	No ATEX, limited Western references
Xingmao	China	1–2 t/h	$300K–$800K	Verified USA/Singapore/Korea installs	After-sales logistics
Henan Doing	China	500–3000 kg/h	$70K–$350K	Transparent pricing, fast delivery	Lowest tier quality risk

The standard process flow needs modernization, not just optimization

The ten-step process (discharge → pre-sort → primary shred → secondary crush → fine milling → screening → magnetic separation → eddy current separation → air classification → black mass collection) reflects **circa-2018 industry thinking**. Current best practice has consolidated this into **5–6 major unit operations** through three key changes.

First, integrated shredding towers combine primary and secondary size reduction into a single unit. The ANDRITZ battery shredding tower (rotary shear + granulator under N_2), ANDRITZ ELDAN TOWER (Super Chopper + Multi Purpose Rasper), Eldan Recycling and BCA Industries' Triplus three-stage system E-Scrap News all achieve ≤15mm output in a single enclosed unit. This eliminates a conveyance step and one piece of standalone equipment, while improving gas containment.

Second, fine milling can often be eliminated. If the shredding tower achieves adequate particle liberation at 10–20mm, and intermediate screening at ~1mm effectively captures liberated black mass powder, a dedicated fine milling stage adds cost without proportionate purity improvement. In fact, over-grinding can *increase* Cu and Al contamination by pulverizing current collector foils into particles too fine to separate from black mass.

Third, the drying step should precede separation rather than follow it. Research from IntechOpen (2024) identifies the dryer — not the shredder — as the rate-limiting bottleneck. Drying before separation dramatically improves downstream separation efficiency because dry particles classify and separate much more predictably than wet, electrolyte-coated material.

A recommended CAPEX-minimized process for 1–5 t/h:

1. **Deep electrical discharge** with energy recovery (Duesenfeld patent, or equivalent)
2. **Integrated shredding tower** under N_2 atmosphere (ANDRITZ, ELDAN, or BCA)
3. **Vacuum drying at <80°C** with electrolyte condensation recovery (Duesenfeld-type)
4. **Screening** at ~1mm cut point (multi-deck vibrating screen)
5. **Separation train**: magnetic → eddy current → air classification
6. **Black mass collection** (cyclone + baghouse)

This configuration uses **6 major units** versus the original 10, eliminates gas scrubbing entirely (if using the Duesenfeld vacuum approach), and creates an electrolyte revenue stream. Pre-sorting is eliminated when feedstock contracts provide chemistry data; fine milling is eliminated because the integrated tower achieves adequate liberation.

Cryogenic and thermal approaches have niche roles, not primary ones

Cryogenic shredding (batteries frozen with liquid nitrogen to −196°C before shredding) has been commercialized by Retriev Technologies and Redivivus in the USA. It effectively suppresses thermal runaway ScienceDirect and can eliminate the pre-discharge step. LinkedIn However, it suffers from **high continuous OPEX** (liquid nitrogen consumption), IntechOpen does not reduce downstream equipment count, and the IntechOpen 2024 review concludes it "faces several limitations for industrial application" IntechOpen at sustained throughput. For a CAPEX-first buyer at 1–5 t/h, cryogenic shredding adds cost without reducing system complexity.

Thermal pre-treatment (pyrolysis at 400–600°C) is more relevant. Pyrolysis decomposes PVDF binder, evaporates electrolyte, and removes separator plastics in one step. ScienceDirect At **580°C, cathode material**

separates from aluminum foil with 84% selectivity, Solarpaces ScienceDirect dramatically improving downstream mechanical liberation. This can reduce or eliminate fine milling and simplify screening. However, pyrolysis generates hazardous off-gases (HF, VOCs, CO_2) requiring scrubbing, and the rotary kiln or retort adds its own significant CAPEX. Suppliers include Accurec (Germany), Genox (China), and SungEel HiTech (South Korea).

Wet/submerged shredding deserves serious consideration. BCA Industries' hydro-nitrogen system BCA Industries RecyclingInside and Shred-Tech's aqueous shredder (based on the Toxco process) combine discharge, shredding, and deactivation into a **single unit operation** Shred-Tech — the most aggressive equipment reduction available. Water suppresses thermal runaway inherently, E-Scrap News and the IntechOpen 2024 review notes it is the "only technique that effectively removes conductive salt from black mass." IntechOpen The trade-off is water treatment infrastructure and the need to dry the black mass before sale. For a buyer in a jurisdiction with less stringent ATEX requirements, wet shredding can meaningfully reduce both CAPEX and safety engineering costs.

Black mass purity from mechanical processing has hard limits

Mechanically produced black mass from end-of-life EV batteries typically contains **1–10% copper, 1–5% aluminum, and 0.5–3% iron** as impurities, alongside 33–50% graphite/carbon and the target cathode metals (Ni, Co, Mn, Li). These impurity levels are a fundamental consequence of incomplete liberation of electrode coatings from current collector foils during crushing. ScienceDirect

The equipment choice with the **highest impact on purity** is multi-stage comminution with intermediate screening. The critical mechanism is fracturing the PVDF binder that adheres cathode/anode active materials to their respective Al and Cu foils, then screening at ~500µm–1mm to capture liberated black mass powder while rejecting coarser foil fragments. A second crushing stage specifically targeting binder adhesion breakage — not just size reduction — is identified in the literature as the **bottleneck of cathode active material recovery**. Air classification is the second most impactful step, separating light foil and separator fragments from denser black mass particles. RecyclingInside Genoxtech

China's new **GB/T 45203-2024 standard** (effective July 2025) sets the first national specifications for recycled black mass. For Grade 1 NMC material, it requires Ni+Co >35%, Li >3.5%, with Cu <0.5%, Al <1%, and Fe <1%. Fastmarkets Meeting these impurity limits mechanically is achievable with production scrap (consistent chemistry, minimal casing contamination) but **extremely challenging with mixed end-of-life batteries**, particularly when steel-cased cells or LFP chemistry are in the feedstock mix. ScienceDirect EOL black mass typically trades at **60–75% payable** against benchmark metal prices, versus 75–95% for production scrap — a direct reflection of purity differences. Benchmarkminerals

Feedstock chemistry matters enormously. **NMC black mass** (Ni 9–25%, Co 5–16%, Li 2–4%) commands the highest market value. M-mines **LFP black mass** contains no nickel or cobalt, making lithium the sole recoverable valuable metal — and the iron from LFP cathodes creates separation nightmares if mixed with NMC material. A buyer processing mixed feedstock should invest in upstream chemistry sorting (XRF-based or similar) to segregate NMC from LFP streams before mechanical processing. Nature

Actionable recommendations for a CAPEX-first buyer

For a buyer prioritizing lower CAPEX at 1–5 t/h scale with mixed EV battery feedstock, the optimal strategy depends on regulatory jurisdiction:

If operating in the EU or other ATEX-regulated environment, the ANDRITZ/Duesenfeld partnership offers the best CAPEX-to-performance ratio among Western suppliers. The elimination of HF gas scrubbing, stainless steel equipment, ANDRITZ and continuous N_2 costs ANDRITZ — combined with electrolyte recovery revenue SlideShare — makes this process fundamentally less expensive than BHS-Sonthofen's conventional approach, despite BHS having more operational references. Budget approximately **€5–12M** for a complete 2 t/h mechanical line through this route. Supplement with Erdwich or ELDAN as alternative quotation sources for competitive tension.

If operating outside strict ATEX jurisdictions (Southeast Asia, Middle East, Africa, parts of Latin America), a Chinese-sourced line from Genox or Xingmao at **$500K–$1.5M** delivers adequate performance for the price. Xingmao's verified 1.5 t/h USA installation and 2 t/h South Korea installation provide the strongest evidence of international viability. Lithiumbatterybroken Budget an additional **15–25%** for installation engineering, commissioning support, and a generous spare parts inventory to mitigate after-sales logistics risk.

For the most aggressive CAPEX minimization regardless of geography, consider wet shredding (BCA Industries or Shred-Tech) to eliminate the discharge station entirely, combined with Chinese-sourced separation equipment. This reduces the total equipment count to approximately **5 major units** and eliminates the most expensive safety engineering around inert gas management.

Regardless of equipment source, invest disproportionately in **screening quality** (multi-deck vibrating screens with precision cut points) and **upstream feedstock sorting** (by chemistry). These two relatively low-cost investments have the highest marginal impact on black mass purity and therefore on the revenue per ton of product — the metric that ultimately determines whether the plant's economics work at any CAPEX level.